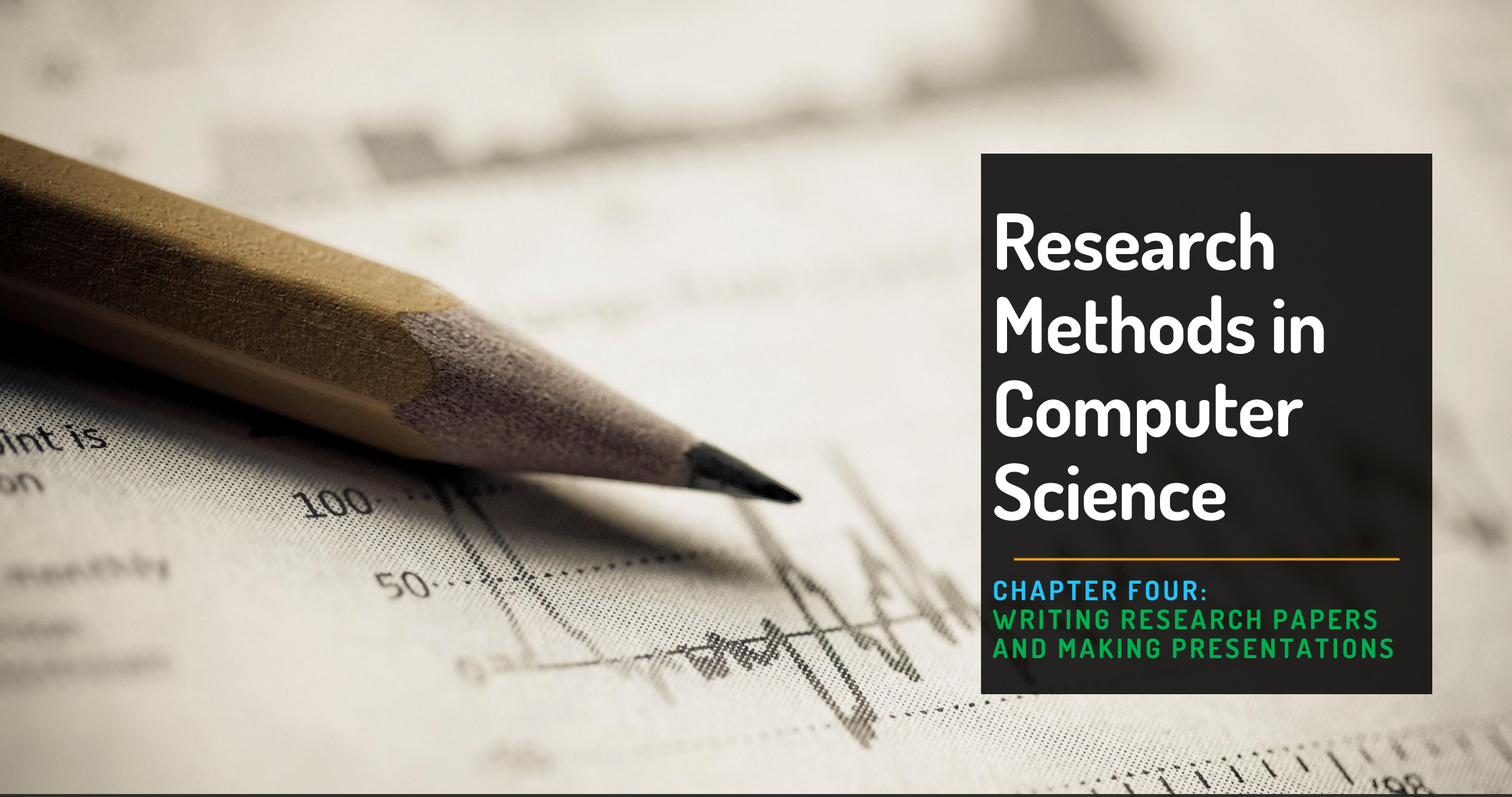

Research Methods in Computer Science

CHAPTER FOUR:
WRITING RESEARCH PAPERS AND MAKING PRESENTATIONS

Contents

- Structure of Good Quality Papers, Citations and References
- Making Excellent Presentation
- How to Write Good Quality Thesis and Papers (Journal and Conferences)

Structure of Good Quality Papers, Citations and References

Why Structure Matters

- Whether you are writing a **B.S.** Degree Research Paper or completing a research report for a certain course, it is highly likely that you will need to organize your research paper in accordance with some structure.
- A typical research paper will contain the following sections in a sequence**(IMRaD**: **(1) Introduction; (2) Methods** and **(3) Results and discussion.)**
- **Benefits:**
  - **For the Reader:** Allows for efficient understanding and critical evaluation.
  - **For the Writer:** Provides a clear template to organize your thoughts.
  - **For the Community:** Ensures consistency and completeness across all scientific literature.

Structure of Good Quality Papers, Citations and References

The IMRaD Structure (The Core)

- **I - Introduction:** What did you study and why?
- **M - Methods:** How did you study it?
- **R - Results:** What did you find?
- **D - Discussion:** What do your findings mean?
- **...And the supporting cast:**
- **Title**
- **Abstract**
- **References**
- **(Optional: Acknowledgments)**

Structure of Good Quality Papers, Citations and References

The Title:

- **Purpose:** To attract the right readers and accurately describe the content.
- **Strategies:**
  - **Descriptive:** ***Example:*** 'A Framework for Efficient Bug Triage in Large-Scale Software Projects'
  - **Results-Oriented:** ***Example***: 'DeepFix: A Fully Convolutional Neural Network for Predicting and Fixing Program Bugs'
- Keep it concise, informative, and include key keywords.

The Abstract:

- **Purpose:** A dense, standalone summary.
- **Structure:** Context & Motivation → Problem/Gap → Method/Approach → Key Results → Conclusion

Structure of Good Quality Papers, Citations and References

Introduction:

- **Goal:** Start broad and funnel down to your specific contribution.
- **Structure:** Broad Area → Specific Problem → The Gap → Your Contribution → Roadmap.

Related Work:

- **Goal:** Show you know the field and position your work relative to others.
- Do NOT just list papers. Synthesize and critique.
- **Bad Example:** 'Author A did X. Author B did Y.'
- **Good Example:** Narrative comparison and critique.

Methodology

- **Goal:** Provide enough detail for replication.
- **Include:** Algorithms, System Architecture, Datasets, Experimental Setup, Implementation Details.

Structure of Good Quality Papers, Citations and References

Results & Discussion

- **Results:** Present facts objectively using figures/tables/graphs.
- **Discussion:** Interpret results, connect to gap, discuss limitations and implications.

Conclusion & References

- **Conclusion:** Restate problem, contributions, key findings, significance.
- **References:** Cite all non-original claims, use consistent style, use reference managers.

Understanding the Different Venues

- The core principles of good writing stay the same, but scope, depth, and audience differ.

Structure of Good Quality Papers, Citations and References

The PhD/Master's Thesis

- **Audience:** Committee evaluating mastery.
- **Purpose:** Demonstrate readiness for research.
- **Characteristics:** Comprehensive, detailed, long (80–200+ pages).

The Journal Paper

- **Audience:** Specialists.
- **Purpose:** Archival, complete scientific contribution.
- **Characteristics:** Rigorous review, extensive experiments, deep discussion.

The Conference Paper

- **Audience:** Active research community.
- **Purpose:** Present timely, novel ideas.
- **Characteristics:** Short, competitive, novelty-focused.

Structure of Good Quality Papers, Citations and References

- Although there are major similarities between multiple research structure templates there are variations in the pattern. **Here are some common research structure templates**:

Research Paper Format | APA, MLA, & Chicago Templates

- The formatting of a research paper is different depending on which style guide you're following. In addition to citations, **APA, MLA,** and **Chicago** provide format guidelines for things like **font choices**, **page layout**, **format** of **headings** and the format of the **reference page**.

Formatting an American Psychological Association (APA) paper

- Use a standard font like **12 pt Times New Roman** or **11 pt Arial.**
- Set **1 inch** page margins.
- Apply **double line** spacing.
- If submitting for publication, insert a running head on every page.
- Indent every new paragraph **½ inch**.

Structure of Good Quality Papers, Citations and References

Formatting a Modern Language Association (MLA) paper: guidelines

- Use an easily readable font like **12 pt Times New Roman.**
- Set **1 inch** page margins.
- Apply **double line** spacing.
- Indent every new paragraph **½ inch.**
- Use **title case capitalization** for **headings.**

Formatting a Chicago paper: guidelines

- Use a standard font like **12 pt Times New Roman.**
- Use **1 inch** margins or larger.
- Apply **double line** spacing.
- Indent every new paragraph **½ inch.**
- Place **page numbers** in the top right or bottom center.

- **More**: https://www.scribbr.com/research-paper/research-paper-format/

Structure of Good Quality Papers, Citations and References

10 helpful tips

1. Focus on a central contribution.
2. Write for those who do not know your work.
3. Use the "context-content-conclusion" approach.
4. Avoid superfluous information and use parallel structures.
5. Summarize your research in the abstract.
6. Explain the importance of your research in the introduction.
7. Explain your results in a logical sequence and support them with figures and tables.
8. Discuss any data gaps and limitations.
9. Allocate your time for the most important sections.
10. Get feedback from colleagues.

Structure of Good Quality Papers, Citations and References

Citations and References

- The terms **reference** and **citation** are also often used to refer to the **same thing** although a **citation** tends to mean the part of the text **within your assignment where you acknowledge the source**; whilst a **reference** usually refers to **the full bibliographic information at the end.**
- The aim of a **citation** is to provide enough **bibliographic information** for the reader to be able to identify and, if necessary, obtain the original resource.
- **Complete**, **correct** and **consistent** citations are very important.
- You may reference a wide variety of resources in your assignment, including **books**, **e-journal articles**, **checklists** and **websites**.
- By using **citations** and **references**, you **acknowledge** the work of others and show how their ideas have contributed to your own work.

Structure of Good Quality Papers, Citations and References

Citations and References

- It is also a way of demonstrating that you have **read** and **understood** **key texts** relating to the area you are writing about.
- The terms **reference list** and **bibliography** are usually used interchangeably, although **strictly** speaking, a **bibliography** refers to **all the reading** you have undertaken for your assignment, **not** just the work you have referred to in your writing.
- When you quote from, or **refer** to, another source of information in your assignment, you must provide a **citation** to it, which then leads to a **reference** giving the full details of the resource.
- You will end up with:
  - a **citation** within the text (**In text citation**)
  - a **reference** in the **bibliography** or **reference list** at or near the end of the assignment (**Bibliography**).

Structure of Good Quality Papers, Citations and References

Citation Style

- A **citation style** dictates the **information necessary** for a **citation** and how the **information** is **ordered**, as well as **punctuation** and **other formatting**.
- There are many different ways of citing resources from your research. The citation style sometimes depends on the academic discipline involved. For example:
  - **APA** (American Psychological Association) is used by Education, Psychology, and Sciences
  - **MLA** (Modern Language Association) style is used by the Humanities
  - **Chicago/Turabian** style is generally used by Business, History, and the Fine Arts
- **More on citation styles: https://pitt.libguides.com/citationhelp**
- **Sample Reference List Citations: https://libguides.dickinson.edu/citing/apasamples**

Structure of Good Quality Papers, Citations and References

Examples (APA)

- **Book**:
  - Author, A. A., & Author, B. B. (Year). *Title of book in italics with only the first word capitalized*. Location: Publisher.
  - Calfee, R. C., & Valencia, R. R. (1991). *APA guide to preparing manuscripts for journal publication. Washington*, DC: American Psychological Association.
- **Scholarly Article**:
  - Author, A. A. (Year). Title of article. *Title of Periodical in Italics, Volume* (Issue), pages.
  - Fuentes, A. (2016). Contemporary evolutionary theory in biological anthropology: Insight into human evolution, genomics and challenges to racialized pseudo-science. Revista Cuicuilco, 23(65), 293-304.
- **Webpage**:
  - Author, A. A. & Author B. B. (Date). Title of page [Format description]. Retrieved from URL
  - National Geographic Society. (2023, March 8). Climate change effects: The consequences of a warming world. National Geographic. https://tinyurl.com/3bw5mh8y
- **More**: https://libguides.dickinson.edu/citing/apasamples

Making Excellent Presentation

- The presentation should include: a **short intro**, your **hypotheses,** a brief **description** of the **methods**, **tables** and/or **graphs** related to your **findings**, and an interpretation of your **data**.
- Here are some tips to help you
- **Timing**. Find out how long your talk should be.
- **Audience**. Find out what sort of audience will listen to your talk.
- **Content**. Students often think they need to **explain every single thing** they know or be perceived as knowing too little. **This is not true.**
- **Organization**. Your talk must have a beginning, middle, and end.
  - introduce yourself; present your research question and why it matters; describe how you conducted your research, explain what you found out and what it means; and conclude with a summary of your main points.

Making Excellent Presentation

A Presentation is Not a Paper

- **Key Difference:** A paper is for study; a presentation is for comprehension.
- **Goal:** Convey the essence and excitement of your work within limited time.

PowerPoint. You should treat PowerPoint as a useful tool.

- **Don't** present too much information on the slides.
- **Do** explain to your audience what each chart or graph indicates.
- **Don't** spend extra time on making a fancy PowerPoint presentation
- **Do** be prepared to give your talk even if technology fails.

- **Tone**. It is best to approach your prepared talk as a somewhat formal occasion.
- **Practice** speaking slowly and clearly.

Making Excellent Presentation

Crafting Your Slide Deck – Structure

The Rule of Three:

- **The Hook:** Title, name, affiliation.
- **The Body:** Introduction & Motivation → Key Idea → Results → Conclusion.
- **The Closing**: Thank You + contact info (kept on screen during Q&A).

Design Principles

- One Idea Per Slide.
- Use Visuals, Not Text.
- Minimize Text: Use short phrases, large fonts (24pt+).
- High Contrast.
- Consistent, Professional Theme.

Making Excellent Presentation

The Art of Delivery

- Know Your Audience.
- Practice with timer; rehearse with peers.
- Manage Nerves: Deep breathing, clarity.
- Engage: Eye contact, slow pace, gestures.
- Q&A: Listen fully, repeat question, respond honestly.

How to Write Good Quality Thesis and Papers (Journal and Conferences)

1. **Most research begins with a question.**
   - Think about which **topics** and **theories** you are interested in and what you would like to know more about.
   - Think about the **topics** and **theories** you have studied in your program.
   - Is there some question you feel the body of knowledge in your field does not answer adequately?
2. **Once you have a question in mind,**
   - begin looking for information relevant to the topic and its theoretical framework.
   - Read everything you can--academic research, trade literature, and information in the popular press and on the Internet.

How to Write Good Quality Thesis and Papers (Journal and Conferences)

3. As you become well-informed about your topic and prior research on the topic, your knowledge should suggest a purpose for your thesis/dissertation.

- When you can articulate this purpose clearly, you are ready to write your prospectus/proposal.
- This document specifies the **purpose of the study, significance of the study, a tentative review of the literature on the topic and its theoretical framework (a working bibliography should be attached), your research questions and/or hypotheses, and how you will collect and analyze your data (your proposed instrumentation should be attached).**

4. At this point, master's students need to recruit committee members (if they haven't done so already) and hold a preliminary meeting.

- The purpose of this meeting is to refine your plans if needed and to make explicit expectations for completion of the thesis.
- At the completion of this meeting, the student should submit a memo to committee members summarizing what was agreed upon during the meeting.

How to Write Good Quality Thesis and Papers (Journal and Conferences)

5. Once your instrumentation is developed, you need to clear it and your informed consent protocol with the Institutional Review Board before you begin collecting data. Leave adequate time to do so. The process can take several days or weeks.

6. Obviously, the next steps are

- collecting and analysing data,
- writing up the findings, and composing the final chapter.
- You also should make sure Chapters 1 and 2 are now fully developed.
- Your chair and committee members provide guidance as needed at this point but expect you to work as independently as possible.

7. Get a copy of the graduate school's guidelines for writing theses and dissertations and follow these guidelines exactly.

Links: https://www.jou.ufl.edu/grad/forms/Guidelines-for-writing-thesis-or-dissertation.pdf
https://www.ldeo.columbia.edu/~martins/sen_sem/thesis_org.html

Universal Principles for Good Writing

- Start Early and Write Regularly.
- Outline First.
- Write for Clarity.
- Get Feedback Early.
- Read Work Aloud.
- Revise Thoroughly.

Chapter Summary & Key Takeaways

- **Research paper =** Story of problem → solution → evidence.
- **Presentation =** Visual highlight of core idea.
- **Know venue differences.**
- **Common thread:** clarity, rigor, contribution.